Radical telescopes for Chebyshev transfers of every degree: normalization, programmed squares, and the fixed-orbit boundary

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This is a non-peer-reviewed companion research note. It consolidates results developed after the quadratic and prime-degree Chebyshev-orbit manuscripts. The transfer is classical and compositional. The proposed contribution is the degree-uniform arithmetic package described below; specialist review of its proofs, sources, and priority remains necessary.

Abstract

For every integer $d \geq 2$, homogenizing T_d gives an integral map from a primitive positive triple $a + b = c$ of opposite parity to another such triple. We identify one fixed Lucas sequence behind the whole orbit. After a necessary normalization at the primes dividing d, the quotient between successive indices d^j is a positive integer $E_{d,j}$. Its cyclotomic indices are precisely

$$\{m : m \mid d^{j+1},\ m \nmid d^j\}.$$

For a natural class of d-admissible seeds, the $E_{d,j}$ are pairwise coprime and disjoint from the seed. This yields the exact all-degree radical telescope

$$\frac{\mathrm{rad}(a_n b_n c_n)}{c_n} = \frac{\mathrm{rad}(a_0 b_0 c_0)}{c_0} \frac{|\sin(d^n\theta)|}{d^n \sin\theta\, W_{d,n}}, \qquad W_{d,n} = \prod_{j<n} \frac{E_{d,j}}{\mathrm{rad}(E_{d,j})}.$$

Composite degree bundles several Lucas atoms in one layer; prime degree is the single-atom case. We also complete the simultaneous finite-genealogy realization for the quadratic transfer and prove a bounded iterated local mean for the logarithmic defect in every prime degree, including 2. For one fixed orbit we isolate an unconditional growing congruence window and show that every fixed valuation depth in a polynomial window is harmless; the remaining obstruction is an exceptionally deep Lucas–Wieferich tail. A dual implementation checks that the canonical quadratic, cubic, and quintic pairs have no new Lucas–Wieferich primes between 10^6 and 10^7, and a 122-digit quadratic layer is completely factored. These results do not prove the fixed-orbit conjecture or any new case of the *abc* conjecture.

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Introduction and novelty boundary

Write

$$\mathrm{rad}(N) = \prod_{p \mid N} p.$$

A primitive positive triple $a + b = c$ is an *abc*-hit when $\operatorname{rad}(abc) < c$. Polynomial identities that transfer one additive triple to another are classical; Martin and Miao survey many examples, and van der Horst studies the transfer method in detail. The Chebyshev identity used here is a homogenized de Moivre identity, not a new polynomial identity.

Lucas atoms and their valuations have also been developed independently of this project. Sagan and Tirrell introduce the atom formalism, while Alecci, Miska, Murru, and Romeo give a general definition and detailed p-adic valuation results. The finite-field factorization behind the local root counts is equivalent, after a Cayley change of variable, to the fibotomic factorization of Byer, Dvorachek, Eckard, Harrington, Wise, and Wong. Ratliff, Rush, and Shah give related radical-preserving cyclotomic constructions.

The proposed contribution is therefore narrower than "a new Chebyshev map." It is the following synthesis:

> one normalization valid for every degree; prime-support separation of its successive multi-atom layers; an exact radical telescope along the resulting additive orbit; constructive control of arbitrary finite square patterns in every prime degree; and a bounded local first moment whose remaining integer-box obstruction is stated explicitly.

Prime degree remains special: a new layer is a single cyclotomic atom (or two coordinate branches of the same order), whereas composite degree groups several atom indices. Since $T_m \circ T_n = T_{mn}$, the composite-degree transfer is itself only a block of lower-degree transfers. The grouping and normalization, rather than the map, are the point.

The last sections distinguish finite-seed averages from a single fixed orbit. They sharpen the fixed-orbit obstruction but do not resolve it. In particular, nothing here should be read as a proof of *abc*.

The universal integral transfer

Let $a + b = c$ be primitive and positive, with a, b of opposite parity, and put

$$D = b - a.$$

Both D and c are odd. Define the homogeneous Chebyshev values

$$H_r(D, c) = c^r T_r(D/c).$$

They are integral because

$$H_0 = 1, \qquad H_1 = D, \qquad H_{r+1} = 2DH_r - c^2 H_{r-1}.$$

Theorem 1 (Universal transfer)**.** *For every integer $d \geq 2$, set*

$$\mathcal{T}_d(a, b, c) = \left(\frac{c^d - H_d(D, c)}{2}, \frac{c^d + H_d(D, c)}{2}, c^d \right). \tag{2.1}$$

Then $\mathcal{T}_d$ maps primitive positive opposite-parity triples to triples of the same kind. If

$$x = \frac{b - a}{c},$$

then the new normalized difference is $T_d(x)$. Moreover

$$\mathcal{T}_m \circ \mathcal{T}_n = \mathcal{T}_{mn} = \mathcal{T}_n \circ \mathcal{T}_m. \tag{2.2}$$

After n iterations,

$$c_n = c_0^{d^n}, \qquad \frac{b_n - a_n}{c_n} = T_{d^n}\left(\frac{b_0 - a_0}{c_0}\right). \tag{2.3}$$

Proof. Since $H_d(D,c)$ is odd, the two coordinates in (2.1) are integers of opposite parity and sum to c^d. The inequality $|T_d(x)| \le 1$ gives nonnegativity. Equality would make $e^{i \arccos x}$ a root of unity, so $2D/c$ would be a rational algebraic integer. This is impossible because $c > 1$, c is odd, and $\gcd(D,c) = 1$. Thus both coordinates are positive.

If an odd prime divided both new coordinates, it would divide c and $H_d(D,c)$. But

$$H_d(D,c) \equiv 2^{d-1} D^d \pmod{c},$$

which is a unit modulo every prime divisor of c. The prime 2 cannot divide both coordinates because their sum is odd. This proves primitivity. The difference of the two new coordinates is $H_d(D,c)$, which proves the semiconjugacy. Equations (2.2) and (2.3) follow from $T_m \circ T_n = T_{mn}$. □

For $d = 2$, (2.1) is exactly

$$(a,b,c) \longmapsto (4ab, (b-a)^2, c^2). \tag{2.4}$$

For odd d, expanding $(\sqrt{b} + i\sqrt{a})^d$ splits the transfer into the familiar two coordinate factors. For even d, write instead

$$(\sqrt{b} + i\sqrt{a})^d = C_d(a,b) + i\sqrt{ab}\, S_d(a,b).$$

Then

$$\mathcal{T}_d(a,b) = \begin{cases} (aS_d(a,b)^2,\ bC_d(a,b)^2), & d \text{ odd}, \\ (abS_d(a,b)^2,\ C_d(a,b)^2), & d \text{ even}. \end{cases} \tag{2.5}$$

Thus every degree has a two-coordinate-factor form, although the seed prefactor depends on parity.

One Lucas sequence and its normalized layers

Fix the seed and put

$$\alpha = D + 2\sqrt{-a_0 b_0}, \qquad \beta = D - 2\sqrt{-a_0 b_0}.$$

Then

$$\alpha + \beta = 2D, \qquad \alpha\beta = c_0^2, \qquad (\alpha - \beta)^2 = -16 a_0 b_0.$$

The associated Lucas sequence is

$$U_0 = 0, \quad U_1 = 1, \quad U_{r+1} = 2DU_r - c_0^2 U_{r-1}, \qquad U_r = \frac{\alpha^r - \beta^r}{\alpha - \beta}. \tag{3.1}$$

Lemma 2 (Fixed-sequence identity)**.** *For the degree-d orbit,*

$$a_n b_n = a_0 b_0\, U_{d^n}^2. \tag{3.2}$$

Proof. The norm identity $c_0^{2d^n} - (\alpha^{d^n} + \beta^{d^n})^2/4 = -(\alpha^{d^n} - \beta^{d^n})^2/4$, together with $(\alpha - \beta)^2 = -16a_0b_0$, gives (3.2). $\square$

Define

$$Q_{d,j} = \frac{U_{d^{j+1}}}{U_{d^j}}. \tag{3.3}$$

Homogeneous cyclotomic factorization gives

$$Q_{d,j} = \prod_{m \in \mathcal{L}_{d,j}} \Phi_m(\alpha, \beta), \qquad \mathcal{L}_{d,j} = \{m : m \mid d^{j+1},\ m \nmid d^j\}. \tag{3.4}$$

The sets $\mathcal{L}_{d,j}$ are pairwise disjoint.

The quotient (3.3) contains a systematic contribution from every prime dividing d. The following seed condition removes it exactly.

Definition 3. A seed is *d-admissible* when it is primitive, positive, and of opposite parity, and for every odd prime $q \mid d$,

$$q \mid a_0, \qquad v_q(U_q) = 1. \tag{3.5}$$

For $q \geq 5$, the second condition follows automatically from the first. At $q = 3$ it is necessary: the end terms in the binomial expression for U_q/q can cancel modulo q.

Theorem 4 (Normalized layer genealogy)**.** *Let $d \geq 2$ and let the seed be d-admissible. Then*

$$E_{d,j} = \frac{|Q_{d,j}|}{d} \tag{3.6}$$

is a positive integer coprime to $da_0b_0c_0$. The integers

$$E_{d,0}, E_{d,1}, E_{d,2}, \ldots$$

are pairwise coprime. Consequently every prime outside the seed support has at most one birth layer.

Proof. For an odd $q \mid d$, the Lucas law of repetition at a discriminant prime, under (3.5), gives

$$v_q(U_r) = v_q(r) \qquad (q \mid r). \tag{3.7}$$

Opposite parity gives the corresponding identity

$$v_2(U_r) = v_2(r) \qquad (2 \mid r). \tag{3.8}$$

These are standard specializations of Lucas-sequence valuation formulas; the hypotheses include $q \nmid \alpha\beta$ and nondegeneracy. Subtracting the valuations at d^{j+1} and d^j gives

$$v_q(Q_{d,j}) = v_q(d) \qquad (q \mid d),$$

so (3.6) is integral and coprime to d.

If an odd $p \mid a_0b_0$, then $U_r \equiv r\alpha^{r-1} \pmod{p}$. This is nonzero when $p \nmid d$, and (3.7) removes the exact systematic part when $p \mid d$. If $p \mid c_0$, one of α, β is zero and the other is a unit modulo p. The parity assertion follows from (3.8) when d is even and directly from the recurrence at odd indices when d is odd. Hence the normalized layers avoid the seed support.

Now suppose $p \nmid da_0b_0c_0$ divided two layers. At a prime above p, a factor $\Phi_m(\alpha, \beta)$ with $p \nmid m$ says that α/β has exact order m; inversion under conjugation does not change that order. The same p cannot therefore occur at indices from two disjoint sets $\mathcal{L}_{d,j}$. This proves pairwise coprimality. $\square$

Remark 5. For an odd prime $d = \ell$, a layer is the single prime-power cyclotomic index ℓ^{j+1}, expressed through two coordinate branches. For $d = 2$, (3.6) is $|b_j - a_j|$. A composite d bundles several atoms in (3.4). Formula (3.6), not a termwise composite-binomial argument, is the stable normalization.

The all-degree radical telescope

Let

$$R_n = \mathrm{rad}(a_n b_n c_n), \qquad W_{d,n} = \prod_{j<n} \frac{E_{d,j}}{\mathrm{rad}(E_{d,j})}. \tag{4.1}$$

Every prime divisor of d already lies in the seed radical: the odd ones divide a_0, and opposite parity handles 2. Telescoping (3.3) and using Theorem 4 therefore gives

$$|U_{d^n}| = d^n \prod_{j<n} E_{d,j}, \qquad R_n = R_0 \prod_{j<n} \mathrm{rad}(E_{d,j}). \tag{4.2}$$

Choose $\theta \in (0, \pi)$ by

$$\cos\theta = \frac{b_0 - a_0}{c_0}.$$

Since $\alpha = c_0 e^{i\theta}$,

$$U_{d^n} = c_0^{d^n - 1} \frac{\sin(d^n\theta)}{\sin\theta}. \tag{4.3}$$

Theorem 6 (All-degree radical telescope)**.** *For every d-admissible seed and every $n \geq 0$,*

$$\boxed{\frac{R_n}{c_n} = \frac{R_0}{c_0} \frac{|\sin(d^n\theta)|}{d^n \sin\theta\, W_{d,n}}.} \tag{4.4}$$

Proof. Divide the two identities in (4.2), substitute (4.3), and use $c_n = c_0^{d^n}$. □

For a fixed algebraic seed, a standard linear-form estimate supplies a constant $C = C(a_0, b_0, d)$ such that $\log|\sin(d^n\theta)| \geq -C(n+1)$; the elementary upper bound is immediate. Thus the exponential-scale obstruction in (4.4) is precisely the powerful part $W_{d,n}$. In the quadratic and prime-degree cases this yields the familiar equivalence

$$\log W_{d,n} = o(d^n) \quad \Longleftrightarrow \quad \text{the orbit quality tends to 1.} \tag{4.5}$$

A persistent quality gap would require $\log W_{d,n}$ to have positive scale d^n. Equation (4.4) is an exact reduction, not a bound on that powerful part.

Programmed squares in prime degree

The prime-degree manuscript proves that any finite compatible collection of birth levels, branches, splitting signs, and exact multiplicities can be realized simultaneously for every odd prime degree. Here we record the missing quadratic counterpart. It is also useful as a source of exact test vectors for the fixed-orbit discussion.

For the quadratic transfer (2.4), let

$$G_n(X,Y) = b_n - a_n$$

be the homogeneous seed-coordinate form and put $m = 2^{n+2}$.

Proposition 7 (Quadratic local roots)**.** *Let p be odd. If $m \mid p - \chi$ for $\chi \in \{1, -1\}$, then $G_n(X,1)$ has exactly 2^n distinct simple roots in $\mathbb{F}_p$. They are*

$$\rho_\zeta = -\left(\frac{\zeta - 1}{\zeta + 1}\right)^2, \tag{5.1}$$

where ζ has exact order m in $\mathbb{F}_p^\times$ when $\chi = 1$, or in the norm-one subgroup of $\mathbb{F}_{p^2}^\times$ when $\chi = -1$, modulo $\zeta \sim \zeta^{-1}$. There are no roots when neither divisibility holds. Every root satisfies

$$\rho \neq 0, -1, \qquad \left(\frac{-\rho}{p}\right) = \chi. \tag{5.2}$$

Each root has a unique Hensel lift, and if $\widehat{\rho}$ is its lift modulo p^h, then for every unit λ,

$$v_p(G_n(\widehat{\rho} + \lambda p^h, 1)) = h. \tag{5.3}$$

Proof. In the quadratic extension generated by square roots of X and Y, the standard Cayley parameter identifies the roots of G_n with exact order-m roots of unity, and inversion has the same seed-coordinate image. The split and norm-one groups have orders $p-1$ and $p+1$, respectively. Degree exhaustion gives $\varphi(m)/2 = 2^n$ roots. The exceptional values $0, -1$ are not roots, and the derivative does not vanish because $p \nmid m$. Formula (5.2) follows directly from (5.1). Hensel's lemma and the first nonzero Taylor term give (5.3). □

Theorem 8 (Simultaneous quadratic realization)**.** *Let*

$$\mathcal{D} = \{(p_i, n_i, h_i, \chi_i) : 1 \leq i \leq r\}$$

have distinct odd primes, $n_i \geq 0$, $h_i \geq 1$, and

$$2^{n_i+2} \mid p_i - \chi_i.$$

There are infinitely many primitive positive seeds with a_0 even and b_0 odd such that, simultaneously for every i,

$$v_{p_i}(E_{2,n_i}) = h_i, \qquad \left(\frac{D_K}{p_i}\right) = \chi_i, \tag{5.4}$$

where $K = \mathbb{Q}(\sqrt{-a_0 b_0})$, while p_i divides neither the seed nor any layer $E_{2,j}$ with $j \neq n_i$.

Proof. For each datum choose a root from Proposition 7 and prescribe its Hensel lift with the perturbation in (5.3) modulo $p_i^{h_i+1}$. Fix $b_0 = 1$. The Chinese remainder theorem, together with $a_0 \equiv 0 \pmod 2$, supplies infinitely many positive a_0 in the resulting progression. These seeds are primitive and have the required parity. Equations (5.2) and (5.3) give the splitting sign and exact valuation. The exclusions $\rho \neq 0, -1$ keep p_i away from $a_0 b_0 c_0$. Finally, the Cayley parameter has exact order 2^{n_i+2}, so it cannot belong to the distinct order class attached to another layer. This proves the last assertion. □

Theorem 8, together with the two-branch odd-prime result, shows that arbitrarily deep but finitely prescribed square behavior is constructible. It does not produce a fixed seed with large defect at infinitely many layers.

A bounded local mean for every prime degree

This section concerns seed averages, not one orbit. Let d be prime. For $d = 2$, let $\mathcal{S}_2(H)$ be the primitive opposite-parity pairs in $[1, H]^2$. Its density is

$$\kappa_2 = \frac{2}{3\zeta(2)}. \tag{6.1}$$

For an odd prime $d = \ell$, take the admissible seed set of the prime-degree manuscript. Its density is

$$\kappa_\ell = \begin{cases} 1/(9\zeta(2)), & \ell = 3, \\ 2/(3(\ell+1)\zeta(2)), & \ell \geq 5. \end{cases} \tag{6.2}$$

Put

$$q_j = \begin{cases} 2^{j+2}, & d = 2, \\ d^{j+1}, & d \text{ odd}, \end{cases} \qquad \nu_d(j) = \begin{cases} \varphi(q_j)/2 = 2^j, & d = 2, \\ \varphi(q_j) = d^j(d-1), & d \text{ odd}. \end{cases} \tag{6.3}$$

The factor $1/2$ in the quadratic case comes from inversion in its single Cayley tower; the odd-prime count combines two branches.

For fixed n, P, K, define the bounded defect

$$D_{n,P,K}(a, b) = \sum_{j<n} \sum_{\substack{p \leq P \\ p \nmid 2d}} \log p \sum_{h=2}^{K+1} \mathbf{1}_{p^h | E_{d,j}(a,b)}. \tag{6.4}$$

Theorem 9 (Bounded iterated local mean)**.** *For every prime degree $d \geq 2$,*

$$\lim_{P\to\infty} \lim_{K\to\infty} \lim_{H\to\infty} \frac{1}{|\mathcal{S}_d(H)|} \sum_{(a,b)\in\mathcal{S}_d(H)} D_{n,P,K}(a, b) = L_d(n), \tag{6.5}$$

where

$$L_d(n) = \sum_{j<n} \nu_d(j) \sum_{\substack{p \nmid 2d \\ q_j | p-1 \text{ or } q_j | p+1}} \frac{\log p}{p^2 - 1}. \tag{6.6}$$

There is an effective constant C_d, independent of n, such that

$$L_d(n) \leq C_d < \infty. \tag{6.7}$$

For odd prime d, one may take

$$C_d = \frac{5}{8} \left[\frac{\zeta(2) \log 3 - \zeta'(2)}{d - 1} + \frac{\zeta(2) d \log d}{(d-1)^2} \right]. \tag{6.8}$$

For $d = 2$, one may take

$$C_2 = \frac{3}{2}\zeta(2) \log 2 - \frac{3}{8}\zeta'(2). \tag{6.9}$$

In particular, $C_d \ll \log d/(d - 1)$ along odd prime degrees.

Proof. At a compatible prime, the relevant level form has exactly $\nu_d(j)$ simple projective roots. Each has one lift modulo p^h and $\varphi(p^h)$ unit representatives. Among the $p^{2h} - p^{2h-2}$ primitive pairs modulo p^h, this gives

$$\Pr(p^h \mid E_{d,j}) = \frac{\nu_d(j)}{p^{h-1}(p+1)}. \tag{6.10}$$

The conditions at 2 and d separate by the Chinese remainder theorem. Summing (6.10) for $2 \leq h \leq K+1$ gives

$$\nu_d(j)\frac{1-p^{-K}}{p^2-1}.$$

For fixed cutoffs the box limit is ordinary counting in finitely many residue classes, which proves (6.5) before the last two limits.

For odd $q \geq 3$, the eligible integers are $2rq \pm 1$, and

$$\sum_{q \mid p^2-1} \frac{\log p}{p^2-1} \leq \frac{5}{8q^2}\left(\zeta(2)\log(3q) - \zeta'(2)\right). \tag{6.11}$$

Multiplying by $\varphi(d^{j+1})$ and summing the two geometric series gives (6.8). When $q = 2^{j+2}$, writing eligible integers as $rq \pm 1$ gives the elementary majorant

$$\sum_{q \mid p^2-1} \frac{\log p}{p^2-1} \leq \frac{3}{q^2}\left(\zeta(2)\log(2q) - \zeta'(2)\right). \tag{6.12}$$

Since $\nu_2(j) = q/4$, summation yields (6.9). □

The theorem deliberately uses the order of limits displayed in (6.5). Passing from this local model to the untruncated integer-box mean would require uniform control of primes growing with the box:

$$\lim_{P\to\infty}\limsup_{H\to\infty} \mathbb{E}_{\mathcal{S}_d(H)}\left[\log W_{d,n} - D_{n,P,\infty}\right] = 0. \tag{6.13}$$

This is a large-square-divisor problem for binary forms. Fixed-modulus Hensel densities do not prove (6.13), and general multivariable results that assume *abc* cannot be used here without circularity.

For composite degrees, (3.4) suggests the convergent majorant

$$\sum_{\substack{m \mid d^\infty \\ m>1}} \frac{\varphi(m)\log(3m)}{m^2}.$$

A theorem would require a fully checked simultaneous root classification for every composite-layer atom. We leave that extension open rather than promote the majorant to a result.

What remains for one fixed orbit

Restrict now to the quadratic orbit or one admissible odd-prime-degree orbit. Write $E_j = E_{d,j}$,

$$\delta_j = \log\frac{E_j}{\operatorname{rad}(E_j)}, \qquad q_j = \begin{cases} 2^{j+2}, & d=2, \\ d^{j+1}, & d \text{ odd}. \end{cases} \tag{7.1}$$

Every sufficiently large prime $p \mid E_j$ satisfies

$$p \equiv \pm 1 \pmod{q_j}. \tag{7.2}$$

Let

$$\mathrm{Sq}(N) = \prod_p p^{\lfloor v_p(N)/2 \rfloor}.$$

Nonnegativity and

$$\log \mathrm{Sq}(E_j) \le \delta_j \le 2 \log \mathrm{Sq}(E_j)$$

give the exact equivalence

$$\log W_{d,n} = o(d^n) \iff \delta_j = o(d^j) \iff \log \mathrm{Sq}(E_j) = o(d^j). \tag{7.3}$$

Thus the fixed-orbit problem is a subpower bound for the largest square dividing one sparse sequence of cyclotomic values.

There is one unconditional growing window. Stewart's valuation lemma for a fixed Lucas pair implies, outside an effective finite set,

$$\mathrm{ord}_{\mathfrak{p}}(u^m - 1) < p \exp\left(-\frac{\log p}{51.9 \log \log p}\right) \log |\alpha| \log m, \tag{7.4}$$

where $u = \alpha/\beta$.

Theorem 10 (A growing harmless window)**.** *Put $L_j = \log q_j / \log \log q_j$. For every fixed $0 < \gamma < 1/103.8$,*

$$\sum_{\substack{p \le q_j \exp(\gamma L_j) \\ p^2 \mid E_j}} (v_p(E_j) - 1) \log p = o(q_j). \tag{7.5}$$

Proof. Choose a' with $2\gamma < a' < 1/51.9$. Since every candidate prime is at least $q_j - 1$, (7.4) contributes a factor at most $e^{-a' L_j}$. There are at most $2(e^{\gamma L_j} + 1)$ candidate integers in (7.2). Multiplying the number of candidates, the window endpoint, the valuation bound, and $\log p$ gives

$$q_j \exp((2\gamma - a' + o(1))L_j) = o(q_j).$$

□

This is larger than every fixed multiple of q_j, but smaller than $q_j^{1+\varepsilon}$ for each fixed $\varepsilon > 0$. Inside such a polynomial window, the exact remaining obstruction can be stated elementarily. For $Y \ge q_j$, put

$$D_j(Y) = \sum_{\substack{p \le Y \\ p^2 \mid E_j}} (v_p(E_j) - 1) \log p.$$

For an integer $K \ge 1$, split it as

$$\begin{aligned} T_{j,K}(Y) &= \sum_{\substack{p \le Y \\ p^2 \mid E_j}} \min\{v_p(E_j) - 1, K\} \log p, \\ R_{j,K}(Y) &= \sum_{\substack{p \le Y \\ p^2 \mid E_j}} (v_p(E_j) - 1 - K)_+ \log p. \end{aligned} \tag{7.6}$$

Theorem 11 (Polynomial window equals its deep tail)**.** *Let $1 \le \beta < 2$, $Y = q_j^{\beta}$, and*

$$K_j = o\left(\frac{q_j^{2-\beta}}{\log q_j}\right).$$

Then

$$T_{j,K_j}(Y) = o(q_j), \qquad D_j(Y) = o(q_j) \iff R_{j,K_j}(Y) = o(q_j). \tag{7.7}$$

Proof. By (7.2), at most $2(Y/q_j + 1)$ candidate integers occur. Therefore

$$T_{j,K_j}(Y) \le 2K_j(Y/q_j + 1)\log Y = o(q_j).$$

The equivalence follows from the nonnegative exact split (7.6). □

In particular, squares, cubes, every fixed valuation depth, and depths growing almost as quickly as $q_j^{2-\beta}/\log q_j$ are harmless. The problem is not merely to show that Lucas–Wieferich primes are rare; it is to rule out an aggregate of extraordinarily deep lifts.

This phrasing is exact. Away from the seed and ramified primes, $p^2 \mid E_j$ forces

$$v_p(U_{p-(\Delta/p)}) = v_p(E_j) \ge 2. \tag{7.8}$$

Thus every squared layer prime is Lucas–Wieferich for the fixed pair. If its valuation is at least 3, the corresponding prime ideal is super-Wieferich. Finiteness of those super-Wieferich prime ideals would, for example, imply $D_j(q_j^{\beta}) = o(q_j)$ for every fixed $\beta < 2$; that finiteness is an open hypothesis, not an available theorem.

Deeper finite computation

Finite computation neither proves nor predicts the fixed-orbit asymptotic. It does give exact regression targets for the layer formulas and the Lucas–Wieferich bridge.

The 122-digit quadratic layer

For the quadratic seed $(a_0, b_0, c_0) = (1, 8, 9)$, the level-seven normalized layer is the 122-digit integer

$$\begin{aligned} E_7 = {} & 78999936752708889867904676370303602782482375124030754597220 95\\ & 78681194181644136499736479707715857952926946734495860112066 57.\end{aligned}$$

GMP-ECM 7.0.7 split the formerly unresolved 98-digit residual by one Pollard $p - 1$ step and two ECM discoveries. Exact division gives

$$\begin{aligned} E_7 = {} & 189439 \cdot 750692351 \cdot 9825841153 \cdot 298196593663\\ & \cdot 991245449894911 \cdot 6726631000961507661177857\\ & \cdot 2843440415162664109113943590903491023744717 3121. \end{aligned} \tag{8.1}$$

All seven factors pass independent primality tests, their exact product is recomputed from the recurrence, each occurs to exponent one, and their residues modulo 512 are

$$511,\ 511,\ 1,\ 511,\ 511,\ 1,\ 1.$$

Thus the forced $p \equiv \pm 1 \pmod{512}$ condition holds and this layer has zero repeated-prime defect.

A census beyond one million

For a Lucas pair (P, Q), the census tests every odd unramified prime $p \le 10^7$ for

$$p^2 \mid U_{p-(\Delta/p)}(P, Q)$$

and then tests the hits modulo p^3. Two implementations were used: binary exponentiation in $\mathbb{Z}[X]/(X^2 - PX + Q)$, and an independently written companion-matrix algorithm. Each scanned all 664,578 odd primes through 10^7, and the outputs agree exactly.

orbit pair (P, Q)	Wieferich p	rank	tower rank?	depth ≥ 3?
quadratic $(14, 81)$	65519	455	no	no
cubic $(-2, 25)$	47	24	no	no
quintic $(-6, 49)$	53	26	no	no

There are no new hits between 10^6 and 10^7. None of the three known hits can enter its displayed orbit tower because its rank is not a pure power of the orbit degree, and none is super-Wieferich. This is a bounded observation only; no independence or density assertion is being made.

The exact inputs, intermediate cofactors, software versions, machine-readable census output, and regression commands accompany the source repository.

Conclusions

The all-degree result is structural. It shows that the quadratic orbit, the odd-prime construction, and every composite block share one normalized Lucas-layer mechanism and one radical identity. The quadratic realization and the prime-degree local mean complete two genuine gaps between the first two manuscripts.

The fixed-orbit result is more modest. Congruence and valuation theory control a growing $q_j^{1+o(1)}$ window, and pure counting disposes of every bounded or moderately growing lift depth in any fixed polynomial window below q_j^2. What survives is a deep diagonal Lucas–Wieferich tail. Neither the complete level-seven factorization nor the census through 10^7 changes that logical boundary.

Accordingly, these papers have moved closer to *abc* in the sense of producing an exact reduction and a sharper partial bound, not in the sense of proving a new case of the conjecture. A next theorem would need genuine uniform control of deep lifts for one fixed Lucas pair, or a different global mechanism for the powerful part.

AI-use statement

OpenAI Codex and Anthropic Claude were used during Phases 5–8 for mathematical exploration, proof and source audits, independent software implementations, computation, drafting, and adversarial cross-review. The named author must independently verify every theorem, proof, reference, novelty statement, and computation before circulation or submission. The AI systems are not authors.

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